

Calculative Foundation Project Report

Matrix Operations

- Matrix addition combines two matrices element-wise.
- Matrix subtraction finds differences between matrices.
- Matrix multiplication follows row-column rule.
- Used in data transformations and ML.
- Requires compatible dimensions.

Examples:

Example 1: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

Example 2: Multiply 2x2 matrices using dot product

Norms

- Norm measures vector length.
- L1 norm = sum of absolute values.
- L2 norm = Euclidean distance.
- Used in ML optimization.
- Helps compare magnitudes.

Examples:

Example 1: $||[3,4]|| = 5$

Example 2: L1 norm of $[1,2,3] = 6$

Eigenvalues & Eigenvectors

- Eigenvalues show variance importance.
- Eigenvectors show direction.
- Used in PCA.
- Helps reduce dimensions.
- Important in data science.

Examples:

Example 1: Solve $Ax = \lambda x$

Example 2: 2x2 matrix eigenvalues calculation

PCA

- Reduces dimensionality.
- Keeps maximum variance.
- Transforms features.
- Removes redundancy.
- Widely used in ML.

Examples:

Example 1: Reduce 3D data to 2D

Example 2: Image compression

LDA

- Separates classes.
- Maximizes class distance.
- Used in classification.
- Requires labeled data.
- Improves accuracy.

Examples:

Example 1: Spam vs non-spam

Example 2: Pass vs fail students